

Already It Was Hard to Tell Who's Speaking Over There, and Now Face Masks! Can Binaural Audio Help Remote Participation in Hybrid Meetings?

Loïc Rosset
contact@loicrosset.ch
Human-IST Institute
University of Fribourg
Fribourg, Switzerland

Sailin Zhong
sailin.zhong@unifr.ch
Human-IST Institute
University of Fribourg
Fribourg, Switzerland

Hamed S Alavi
hamed.alavi@unifr.ch
Human-IST Institute
University of Fribourg
Fribourg, Switzerland

Denis Lalanne
denis.lalanne@unifr.ch
Human-IST Institute
University of Fribourg
Fribourg, Switzerland

ABSTRACT

In the context of office-work meetings, it has become a norm that one or more participants attend remotely while others are in the (physical) meeting room—the social situation that has been studied as “hybrid meetings”. We examine whether incorporating the direction of sound in the audio can support the remote attendees to recognize more clearly who is speaking in the meeting room and eventually improve the experience of attending a hybrid meeting. We present the results of a user study, in which 42 participants followed six different discussions recorded in a meeting room, in six conditions: three audio formats are examined, once in a situation where the co-located conferees wore a face-mask and once without a mask. The results demonstrate that the binaural audio can support remote participation, especially in terms of general comprehension and confidence of comprehension, with higher effect for the face-mask conditions.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *Empirical studies in interaction design*; Empirical studies in collaborative and social computing.

KEYWORDS

Video-Conferencing, Hybrid Meeting, Home-Office, Binaural Sound

ACM Reference Format:

Loïc Rosset, Hamed S Alavi, Sailin Zhong, and Denis Lalanne. 2021. Already It Was Hard to Tell Who's Speaking Over There, and Now Face Masks! Can Binaural Audio Help Remote Participation in Hybrid Meetings?. In *CHI '21 Extended Abstracts*, May 8–13, 2021, Yokohama, Japan

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451802>

Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451802>

1 INTRODUCTION

Teleconferencing plays an increasingly important role in office-work practices due to a wide span of reasons including the geographical diversity of collaborators, its benefits on carbon footprint, as well as health-related recommendations such as those associated with the COVID-19 pandemic. Correspondingly, the HCI research community has offered notable contributions to the design of on-line meeting apparatus and analysis of social interactions in such contexts (e.g. [5, 7, 14, 27]). One of the meeting models that is foreseen to continue thriving even after the pandemic is the hybrid model, i.e. a mixture of co-located and remote participation. It is estimated that in 2020, a quarter of the meetings were held in hybrid model [3]. In addition, several studies indicate that working from distance will remain as a regular practice. For example, a recent study of the pandemic's effect on the future of work, conducted by IBM, revealed that 54% of adults would like home to remain as their primary work environment [12], and a study by Global Workplace Analytics predicts that 25-30% of the workforce will be working remotely multiple days a week by the end of 2021 [16].

Despite technological advances in the last 30 years to improve the hardware capabilities including camera, microphone, speakers for high quality audio and video (e.g. [17, 28, 30]) as well as software telecommunication platforms, it has been recognized that hybrid meetings are yet far from providing an impartially accessible, engaging, and undisrupted experience for the remote and co-located conferees [22]. The contribution that we present in this paper investigates the experience of remotely attending a hybrid meeting, with a particularly focus on the features of the audio communication.

The necessity to improve auditory features in videoconferencing have been recognized since a long time ago. The iconic Media Space project created in the Xerox PARC lab in the mid-1980s already acknowledged that discriminating speakers is essential to gain conferees' attention in group activities like brainstorming, but it is difficult to be achieved by showing the group audio and video

without separating it with respect to the in-person conferees [5]. Later research also verified the importance of speaker identification and introduced it as one of the top technical problems [31, 32] and a fundamental component for establishing social micro-connection with other meeting participants [15, 23].

The specific goal of this contribution is to examine the potential of binaural audio to help identify the (co-located) speaker, and thus improve the experience of remote participation in hybrid meetings. The technical goal is that the remote conferees perceive the sound from the direction where the speaker is located in the meeting room. This is done using binaural audio and simulated via the two channels (left and right) of a headphone/earphone. We present a user study to evaluate the impact of binaural audio on the experience of remotely following a discussion, in terms of general comprehension, memory, and perceived helpfulness. For clarity, we would like to emphasize that the technique that we use – binaural audio – has the capacity to transmit the azimuth, elevation and direction of the source naturally with two channels in three dimensions (contrary to stereo) as the auditory system of humans does, in contrast to methods that merely try to recreate this immersive effect with more speakers or channels (e.g. surround sound, ambisonic).

2 RELATED WORK

In this section, we briefly review some of the previous related works that have contributed to two domains: (1) the user experience of hybrid meetings and (2) speaker identification using audio features.

Asymmetric Engagement in Hybrid Meetings. Despite the remarkable diversity in formats and social constructs of hybrid meetings in different working environments, ethnographic studies have revealed common problems including practical difficulties in setting up the hardware and software platforms and disruptions due to technical hazards (e.g. [18, 22, 26]), but also of unbalanced participation and asymmetries of social and cultural context across the co-located and remote conferees [23]. Consequently, videoconferencing tools have been proposed that empower the remote participants. Akker et al. proposed a system to improve remote engagement and control over the discussion floor [18]. They have employed modules that can recognize the visual focus of attention for each participant, speech recognition techniques, as well as a module that can notify who is being addressed by the speaker. By making such information visible to the participants they created a system that allowed for higher engagement of remote attendees. Xu et al. proposed a different approach in which the videoconferencing tool amplifies the visibility of the remote participants [29]. By highlighting where the remote participants' gaze is directed, the local participants may acknowledge better their points of interest and thus change their behavior towards more balanced opportunities for participation. The latter is the eventual objective rendered for many of the contributions that seek to scaffold the social construct in hybrid meeting model, and similarly for the study presented in this paper.

Speaker Identification with Audio. As humans we are capable of selecting which auditory information we want to listen to in noisy environment, a capacity known as the cocktail party effect [19]. In the mentioned case of identifying who is speaking among multiple speakers, Hawley and Litovsky showed that the benefit of binaural hearing, generated virtually with head-related impulse response, is particularly pronounced for speech intelligibility [9].

In Human-Computer Interaction, several studies have been made to evaluate the benefits of spatial audio (e.g. [4, 10, 11]). A stepping stone, is the study of Baldis [2], who demonstrated that in a recorded audio-conference with 4 conferees, participants experiencing spatial audio reproduced on surround speakers, had their focal assurance, their perceived comprehension and their memory improved compared to non-spatial audio. These benefits have been reproduced with binaural audio in a similar study with a listening task by Kilgore and Chignell [13], who found a significant effect on memory when participants had the possibility to choose the location of conferees. When testing remote conferencing systems, it is important to know that a listening task does not produce the same results as a conversational task. Indeed, it seems that listening tasks use more cognitive load than conversational tasks [25]. Moreover, Raake and Wierstorf mentions that the internal references and expertise of the participants, which determine the level of their mental models regarding a video-conferencing system or audio quality, will also influence their judgment of the quality of experience (QoE) [21]. This is true as well for the socio-cultural background of the participants and the context of experience. For example, in a study they made on spatial audio conferencing, when comparing their study which includes 3 conferees, with the one of Baldis [2] which includes 4 of them, they point at the fact that having more conferees to identify and distinguished between, leads to an increase usefulness of spatial audio [1].

While these results are inspiring, they were obtained on audio-conferences. Inkpen et al. explored the impact of spatial audio and video in remote conversations [11]. They found that adding videos of the remote conferees made a significant improvement in the distributed conversations, but interestingly, that spatial audio did not make any difference. In their study, they used a low-fidelity representation of spatial sound, with two speakers positioned on the left and right of a widescreen monitor. Moreover, the participants were actively participating in a conversation with two remote persons, which contributes to a lower cognitive load. Thus, these results do not show if a concrete spatial audio, like binaural audio, do make a difference in video-conferences with co-located conferees, who could even wear masks because of the Covid context. Therefore, the study conducted in this article aims at answering this question.

3 STUDY OBJECTIVES

The study that we present in this paper aims to examine the opportunities that advances in sound specialization techniques can create to improve inter-personal communication in the context of office meetings where some of the conferees attend remotely¹. More specifically, we seek to answer whether the binaural audio streaming of meeting room conversation to the remote conferees (compared to mono audio) can have an impact on their experience of video-conferencing in the following terms: (RQ1) general comprehension of discourses in the meeting room, (RQ2) the feeling of presence and engagement, (RQ3) memory of who has spoken about what subjects, and (RQ4) the user experience of video-conferencing

¹Throughout this article, the terms “co-located conferees or attendees” refer to the individuals who take part in a video-conference and are physically together in the same room. They are normally recorded with only one camera and one microphone. On the other hand, “remote conferees” take part in the video-conference remotely and use a personal camera and microphone.

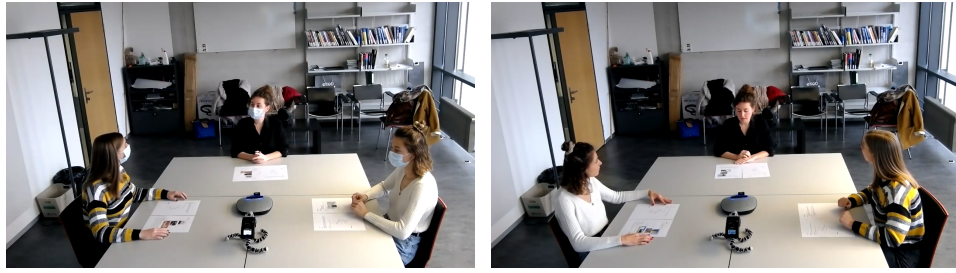


Figure 1: Screenshots of recorded conferences with conferees in the two conditions: with and without a surgical face mask. Three female actors were recruited and instructed to play the role of co-located conferees.

tool. Conjecturing that in situations where the co-located conferees wear a face mask the task of following the meeting discussion may be particularly more challenging for the remote conferees, the above research questions will be investigated both in settings where the co-located conferees do or do not wear a mask. In addition, to investigate the effects of face visibility, we seek to answer (RQ5) what impacts wearing masks in the meeting room can have on the experience of remote conferees.

4 EXPERIMENTAL DESIGN

The research questions we outlined in the last section bring about two independent variables: 1) *auditive format* including (a) “mono”, which is currently used by popular video-conferencing systems, (b) “binaural” to carry the direction of sound, and (c) “binaural-with-head-rotation”, in which we consider the assumption that the remote conferee’s head rotates, after a short while, towards the co-located conferee who just started speaking (further explanations in section 4.1), 2) *face visibility* including the situations where co-located conferees wear or do not wear masks. In order to simulate similar conferences across the conditions, six semi-controlled structured dialogues (see Figure 2) were created as video recordings by three actors who played the role of co-located conferees. The videos were presented to the study participants (as remote conferees) in a counterbalanced order embedded in an online survey portal. In the reminder of this section, we will first describe the video production process and then explain how the experiment was conducted.

4.1 Video production of co-located discussion

Recruited actors. Three caucasian women aged between 21 and 22 years old, with same skin color acted as three co-located conferees. They were students at the University of Fribourg and did not participate in the study (no overlap between the participants of the study and the participants in the video production).

Discussion Structure. To generate the discussions, we adopted the structure that Raake and Schlegel proposed [20] (see Figure 2) in a semi-controlled manner, i.e. the actors did not receive a fixed script; instead they received instructions about the structure of discussion along with information that they used during the discussion. The information sheet for each actor was different and included unique information that the other actors did not have. Six

topics were discussed with this structure and recorded: (1) organizing an enterprise meeting, (2) finding a title for a music album, (3) organizing a travel, (4) organizing an art exhibition, (5) finding a design for the renovation of a bathroom, and (6) organizing a banquet. The total duration of each video is around 2 minutes. For each video/topic the three actors changed their place around the table and their role, in order to avoid learning effects. Three discussions were recorded with face mask and three without mask. The audio of two recordings – one with and one without mask – were kept in mono format. Two transformed to binaural, and two transformed to binaural-with-head-rotation.

Audio Post-processing. The binaural audio was manually rendered using head-related transfer functions (HRTFs) with Sound-scape renderer (SSR) [8] where the mono recording was manually placed around the listener with regards of the speaker location. For the binaural-with-head-rotation, previously rendered binaural source was used, with a rotation of the audio after approximately half a second each time a co-located conferee spoke. This was done such that the speaking co-located conferee acoustically appeared at a location around the head of the remote conferee, and then, was rotated back to the front (if not already there), mimicking a head rotation of the remote conferee. This rotation was also done manually, which may be significantly improved using head-tracking or eye-tracking methods in future works.

4.2 Experimental procedure

The six videos were arranged with Latin square counterbalancing to each experiment participant. The experiment used within-group design to reduce the impact on memory and attention caused by individual differences.

User Participation. We set our objective to have 42 participants to be able to achieve above 80% statistical power with large effect size for the independent variables. The participants are mostly students of the University of Fribourg in Switzerland and recruited through mailing lists and social media. Descriptive information of the participants such as their age, gender and frequency in attending remote meetings are described in section 5. As compensation, the participants were offered the chance (randomly selected) to win one of the five vouchers (each around 100 USD).

Surveying. The survey, inspired by the experiment on spatial audio by Baldis [2], contained (1) a pre-experimental questionnaire on

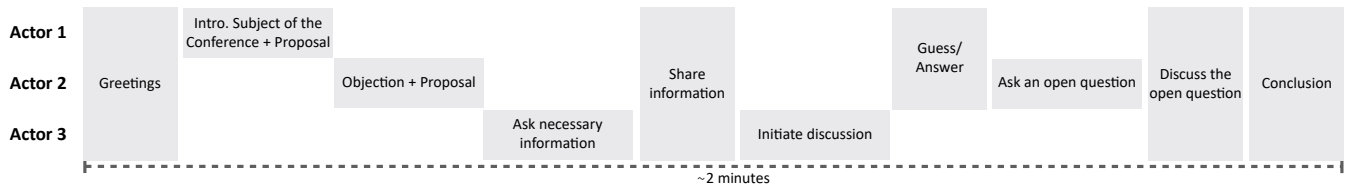


Figure 2: Three actors discussed six topics for the video recordings of co-located conferees in the hybrid meeting. The structure of the discussions, as illustrated in this figure, is inspired and adopted from the work of Raake and Schlegel [20].

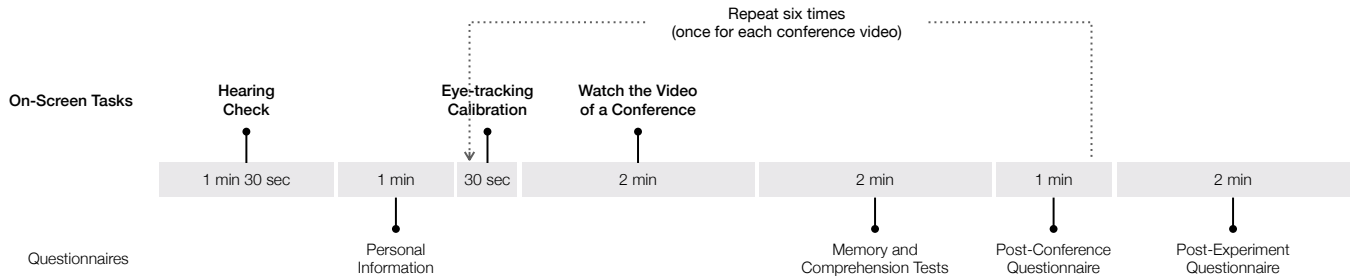


Figure 3: Experimental design of the study, illustrating what was requested from the study participants to follow. The indicated time durations are approximative.

demographic information, (2) objective measurements of comprehension and memory using tests as well as subjective measurements of confidence via self-reported questionnaires for each video, and finally (3) a post-experimental questionnaire (see Figure 3). The participants were briefed about the task and about the questionnaires (without any mention of the use of binaural audio) at the beginning of the survey.

Before watching the first video, two steps related to technical setup were required. the participants were asked to test their headphone or earphone. The *auditory test* consisted of listening to a recorded voice which came from three different direction (front, left, right) using a binaural audio and the participants had to tell from which direction they were able to recognize it.

The *objective measurements* included two tests: a memory test and a comprehension test. The memory test showed pictures of the co-located conferees (with or without mask depending on the condition) and six statements which were needed to be linked to the corresponding co-located conferee. The comprehension test consisted of three multiple choice questions based on the subject, as well as the raised issues and the decisions of the conferees. The *subjective measurements* were conducted in a form of rating in a five point Likert scale, including two questions right after the memory test and comprehension test separately to inquire the self-reported confidence on their answers. After each video we also asked the participants about speaker identification difficulty, overall comprehension difficulty, concentration needed, feeling of presence, and audio helpfulness. The *post-experimental questionnaire* asked the participants about their potential strategy to recall who said what in the conference and about their awareness of the binaural sound in the videos. The participant also needed to rate how the binaural sound impacted their comprehension of the discussions

and say if it was distracting or not. An open question was finally provided for any feedback or comment from the participants.

5 ANALYSIS AND RESULTS

We collected data from 42 participants who completed all the steps among which 16 identified as female, 22 male, and 4 unidentified. A majority (39) were aged between 18 and 25 years old and 3 between 26 and 34 years old. Most participants (37) had used video-conferencing systems more than once a week. All participants distinguished the direction of sound wearing headphones or earphones with 37 who had a perfect audition according to our auditory test.

For each dependant variable measured in the study, a test of the normal distribution of the data and of the homogeneity of the variances are made. It resulted that all of the variables violated one or more assumptions required for parametric statistical tests. Therefore, non-parametric Friedman's tests of differences among the repeated measures are conducted for all the dependent variables: comprehension, comprehension confidence (subjective), memory, memory confidence (subjective), speaker identification difficulty (subjective), feeling of presence (subjective), audio helpfulness (subjective). All the statistical test are done for the mask and without-mask conditions separately. In each test, the three auditive formats were compared resulting in a degree of freedom of 2. The results of the Friedman's tests allowed to identify significant differences among the auditive formats and post-hoc analysis with Wilcoxon signed-rank tests were conducted to determine the significance of difference between each pair of auditive formats. The results of Wilcoxon signed-rank tests are summarized in table 1. In the end, an analysis of the interaction effects between the visual conditions with and without face masks was performed with Friedman's tests and post-hoc Wilcoxon signed-rank tests.

Table 1: Table showing the post-hoc Wilcoxon signed-rank tests between auditive format 1 and 2 and for each of the dependant variables which were significant on the Friedman's tests. The effect size (r) was calculated with Pearson's correlation coefficient and the p values have been corrected for multiple tests with Bonferroni correction. Beside each auditive format lies the obtained mean and the best condition in each significant test is displayed in bold.

	Dependant Variable	Auditive Format 1	Auditive Format 2	Z-Score	p-value	Effect size (r)
Without Mask	Comprehension score (Min. = 0, Max = 3)	Binaural (M=2.79)	Mono (M=2.26)	-3.554	0.001 ***	-0.548
		Binaural rotation (M=2.81)	Mono (M=2.26)	-3.518	0.001 ***	-0.543
		Binaural rotation (M=2.81)	Binaural (M=2.79)	-0.237	2.439	-0.036
	Comprehension confidence [subjective] (Likert's scale: 1=very low, 5=very high)	Binaural (M=3.60)	Mono (M=3.62)	-0.059	2.859	-0.009
		Binaural rotation (M=4.24)	Mono (M=3.62)	-2.925	0.009 **	-0.451
		Binaural rotation (M=4.24)	Binaural (M=3.60)	-3.028	0.006 **	-0.467
	Speaker identification difficulty [subjective] (Likert's scale: 1=very difficult, 5=very easy)	Binaural (M=4.45)	Mono (M=4.26)	-1.413	0.474	-0.218
		Binaural rotation (M=4.00)	Mono (M=4.26)	-1.507	0.396	-0.233
		Binaural rotation (M=4.00)	Binaural (M=4.45)	-2.559	0.03 *	-0.395
	Audio helpfulness [subjective] (Likert's scale: 1=very unhelpful, 5= very helpful)	Binaural (M=3.29)	Mono (M=3.40)	-0.417	2.031	-0.064
Binaural rotation (M=2.76)		Mono (M=3.40)	-2.935	0.009 **	-0.453	
Binaural rotation (M=2.76)		Binaural (M=3.29)	-2.194	0.084	-0.338	
With Mask	Comprehension score (Min. = 0, Max = 3)	Binaural (M=2.19)	Mono (M=2.43)	-1.447	0.444	-0.223
		Binaural rotation (M=2.90)	Mono (M=2.43)	-3.123	0.006 **	-0.482
		Binaural rotation (M=2.90)	Binaural (M=2.19)	-4.259	< 0.001 ***	-0.657
	Comprehension confidence [subjective] (Likert's scale: 1=very low, 5=very high)	Binaural (M=3.00)	Mono (M=3.67)	-2.647	0.024 *	-0.408
		Binaural rotation (M=4.14)	Mono (M=3.67)	-1.829	0.201	-0.282
		Binaural rotation (M=4.14)	Binaural (M=3.00)	-3.91	< 0.001 ***	-0.603
	Speaker identification difficulty [subjective] (Likert's scale: 1=very difficult, 5=very easy)	Binaural (M=3.64)	Mono (M=3.14)	-1.841	0.198	-0.284
		Binaural rotation (M=3.02)	Mono (M=3.14)	-0.542	1.764	-0.084
		Binaural rotation (M=3.02)	Binaural (M=3.64)	-2.253	0.072	-0.348
	Audio helpfulness [subjective] (Likert's scale: 1=very unhelpful, 5= very helpful)	Binaural (M=3.62)	Mono (M=3.26)	-1.52	0.387	-0.235
		Binaural rotation (M=2.90)	Mono (M=3.26)	-1.724	0.255	-0.266
		Binaural rotation (M=2.90)	Binaural (M=3.62)	-2.566	0.03 *	-0.396
Memory confidence [subjective] (Likert's scale: 1=very low, 5=very high)	Binaural (M=3.02)	Mono (M=3.31)	-1.18	0.714	-0.182	
	Binaural rotation (M=3.48)	Mono (M=3.31)	-0.699	1.455	-0.108	
	Binaural rotation (M=3.48)	Binaural (M=3.02)	-1.674	0.282	-0.258	

5.1 Answering research questions

(RQ1) *General comprehension.* The results of the quantities analysis suggest a positive answer. Specifically, for the videos where co-located conferees did not wear masks, the average comprehension score was significantly higher in binaural and binaural-with-head-rotation conditions compared to mono. Also the participants reported significantly higher confidence of their comprehension in binaural-with-head-rotation condition compared to the other two auditive formats. For videos where co-located conferees did wear masks, the average comprehension score was significantly higher in binaural-with-head-rotation condition compared to the other conditions. The mean self reported confidence of comprehension in binaural-with-head-rotation conditions is higher than the two other conditions (however, the difference between binaural-with-head-rotation and mono conditions cannot be reported as significant $p=0.201$).

(RQ2) *Feeling of presence and engagement.* There was no statistical difference between the auditive formats concerning the participants feeling of presence to the conferences. The median of the subjective rating of presence for every conditions was at 3 (neutral) except for the binaural condition which was at 4 (present) when no face masks were used; and with masks the median value for all the auditive formats were at 2.5 (between not present and neutral).

(RQ3) *Memory.* There was no significant differences in the score of the memory test or in the confidence of answers to it, when comparing the auditive formats in either with or without mask situations.

(RQ4) *User experience.* Concerning the average rating of “audio helpfulness”, the binaural-with-head-rotation received the lowest score both in the mask and no-mask situations. Similar results

achieved with the “speaker identification difficulty” variable. No significant difference was found, for either of these variables, between mono and both binaural formats. All of the participants (with one exception) recognized the spatial audio produced by the binaural conditions, and the subjective helpfulness rating of it in the final questionnaire led to a median of 4, meaning that they reported the spatialization of audio was being helpful. However, when asking the participants to answer yes (1) or no (0) if the spatial audio was distracting in the final questionnaire, it resulted a mean of .55 leading to a median of 1, meaning that beside being helpful it was also distracting. This might be because of the unfamiliarity of this type of audio for the participants – the fact that also appeared in some of the comments and provided by them (translated to english): P2 “*The first video with spatial sound was difficult to follow. In the other videos, the spatial sound was useful (to locate the person speaking), but perhaps tiring.*”; P34 “*The spatial sound was not disturbing as long as the interlocutors were visible (without masks).*[...]” P42 “*The spatial sound in videos where the speakers wear masks is very useful and makes tracking as easy as if they did not wear masks.*”

There was definitely a mixed feeling about the spatial audio produced by the binaural audio among the participants. This might be due to the fact that the functions used to produce the binaural audio and the head-rotation feature are generic and were not personalized depending on the hearing preferences and eye movement behavior. Also when we asked the participants if they had a particular strategy to remember who said what and to identify who was speaking they responded with actions, some of which do not depend on the audio. Some example responses are as followings: P5 “*I focused on the colours of the clothes of the people speaking and their place in the room.*”; P11 “*I relied on my visual memory.*”; P34 “*I focus mainly*

on listening and voices... I was completely indifferent to “seeing” the video-conference.”

(RQ5) *Mask Effect.* There was an interaction effect between the visual conditions with and without masks in the memory test score. Indeed, the mono ($M=4.64$, $SD=1.428$) and binaural-with-head-rotation ($M=4.76$, $SD=1.411$) auditive formats had a mean memory score higher in mask condition compared to without mask: mono ($M=4.05$, $SD=1.306$), head ($M=4.07$, $SD=1.237$). Also, as expected, in all auditive conditions, the participants reported more difficulty in speaker identification and higher need for concentration when the co-located conferees wore masks.

6 DISCUSSION AND CONCLUDING REMARKS

The results of the presented study suggest that incorporating the direction of sound in the teleconferencing audio can support remote participants in following the meeting room conversation, understand more clearly the nuances of the discussion, and be more confident about their understanding. These, consequently, can result in approaching an optimal situation where equal opportunities for engagement may be available to the remote and co-located attendees of a hybrid meeting. Three discussion points about this contribution are worth noting, as following:

Practical Impact. Unlike professional binaural recording systems that are highly costly and may entail operational expertise, the method that we have used functions on any microphone available in meeting rooms. It necessitate a (mono) sound signal transmitted from the meeting room, along with time-stamped data about the direction of sound, which would be simulated using binaural audio through the headphone channels. On the other hand, the state of technologies that can capture the direction of the sound in a room at real-time has progressed remarkably within the last decade. Machine learning models have been trained with the echoing signature that any physical-casing of a microphone generates – reaching a solution that can estimate the direction of sound using only one simple microphone [6, 24].

Limitations. (a) The participants of our study (remote conferees) did not have a chance to interact with the actors (co-located conferees) and the experimental design did not allow for evaluating the impact of our intervention on user engagement. We admit this as a limitation that future research with in-the-wild approach may rectify. However, it has been argued that the experience of remote participation in following a meeting-room conversation is a particularly challenging task and requires more cognitive load than when speaking, resulting in less attention to the audio. (b) The functions applied on the sound signal for the simulation of direction (HRTFs) are generic functions, while hearing is a subjective experience. Therefore the binaural effect might appear more natural to some of the users. (c) The six conditions and the corresponding questions are designed to be very similar in terms of level of complexity, but has not been formally verified. In the future we plan to conduct another user experiment to formally test their similarity.

Future Research. The envisioned universal move towards remote work has been accelerated in recent years, first due to the environmental concerns and then leaped with the COVID-19 lockdown, leading to the broad recognition that the future of work hinges on the experience of cooperation from distance. Audio is one of the

aspects that we believe – given the results of our study – can bring breakthrough improvement to the teleconferencing experience, but also to every other areas where spatial cues in the audio output can improve the user experience (e.g. news, radio), and thus we argue for broadening design research investigations in that direction. In the continuation of this research, in particular, we will seek to gain insight into the users’ strategies to follow a hybrid meeting and how it may be impacted by the auditory features, for example by incorporating the analysis of users’ gaze data and more extensive qualitative research.

REFERENCES

- [1] Jens Ahrens, Matthias Geier, Alexander Raake, and Claudia Schlegel. 2010. Listening and conversational quality of spatial audio conferencing. In *Audio Engineering Society Conference: 40th International Conference: Spatial Audio: Sense the Sound of Space*. Audio Engineering Society, Deutsche Telekom Laboratories, Technische Universität Berlin, Berlin, Germany, 4–7.
- [2] Jessica J. Baldi. 2001. Effects of spatial audio on memory, comprehension, and preference during desktop conferences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '01)*. Association for Computing Machinery, New York, NY, USA, 166–173. <https://doi.org/10.1145/365024.365092>
- [3] barco.com. 2020. *What are hybrid meetings? Why should you prepare your offices for them?* Barco. <https://www.barco.com/en/news/2020-11-03-what-are-hybrid-meetings>
- [4] Mark Billinghurst, Jerry Bowskill, Mark Jessop, and Jason Morphet. 1998. A wearable spatial conferencing space. In *Digest of Papers. Second International Symposium on Wearable Computers (Cat. No. 98EX215)*. IEEE, Pittsburgh, PA, USA, 76–83.
- [5] Sara A. Bly, Steve R. Harrison, and Susan Irwin. 1993. Media spaces: bringing people together in a video, audio, and computing environment. *Commun. ACM* 36, 1 (Jan. 1993), 28–46. <https://doi.org/10.1145/151233.151235>
- [6] Dalia El Badawy, Ivan Dokmanić, and Martin Vetterli. 2017. Acoustic DoA Estimation by One Unsophisticated Sensor. In *Latent Variable Analysis and Signal Separation*, Petr Tichavský, Massoud Babaie-Zadeh, Olivier J.J. Michel, and Nadège Thirion-Moreau (Eds.). Springer International Publishing, Cham, 89–98.
- [7] Robert S. Fish, Robert E. Kraut, Robert W. Root, and Ronald E. Rice. 1992. Evaluating Video as a Technology for Informal Communication. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Monterey, California, USA) (CHI '92)*. Association for Computing Machinery, New York, NY, USA, 37–48. <https://doi.org/10.1145/142750.142755>
- [8] Matthias Geier, Jens Ahrens, and Sascha Spors. 2008. The SoundScape Renderer: A unified spatial audio reproduction framework for arbitrary rendering methods. In *In 124th AES Conv.* Citeseer, Deutsche Telekom Laboratories/Technische Universität Berlin.
- [9] Monica L. Hawley, Ruth Y. Litovsky, and John F. Culling. 2004. The benefit of binaural hearing in a cocktail party: Effect of location and type of interferer. *The Journal of the Acoustical Society of America* 115, 2 (Jan. 2004), 833–843. <https://doi.org/10.1121/1.1639908>
- [10] Mansoor Hyder, Michael Haun, and Christian Hoene. 2010. Placing the participants of a spatial audio conference call. In *2010 7th IEEE Consumer Communications and Networking Conference*. IEEE, Las Vegas, NV, USA, 1–7.
- [11] Kori Inkpen, Rajesh Hegde, Mary Czerwinski, and Zhengyou Zhang. 2010. Exploring spatialized audio & video for distributed conversations. In *Proceedings of the 2010 ACM conference on Computer supported cooperative work (CSCW '10)*. Association for Computing Machinery, New York, NY, USA, 95–98. <https://doi.org/10.1145/1718918.1718936>
- [12] Jeannine Kilbride. 2020. *IBM Study: COVID-19 Is Significantly Altering U.S. Consumer Behavior and Plans Post-Crisis*. IBM. <https://newsroom.ibm.com/2020-05-01-IBM-Study-COVID-19-Is-Significantly-Altering-U-S-Consumer-Behavior-and-Plans-Post-Crisis>
- [13] Ryan Kilgore, Mark Chignell, and Paul Smith. 2003. Spatialized Audioconferencing: What Are the Benefits?. In *Proceedings of the 2003 Conference of the Centre for Advanced Studies on Collaborative Research (CASCON '03)*. IBM Press, Toronto, Ontario, Canada, 135–144.
- [14] Benjamin Koehne, Patrick C. Shih, and Judith S. Olson. 2012. Remote and Alone: Coping with Being the Remote Member on the Team. In *Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work (Seattle, Washington, USA) (CSCW '12)*. Association for Computing Machinery, New York, NY, USA, 1257–1266. <https://doi.org/10.1145/2145204.2145393>
- [15] Anastasia Kuzminykh and Sean Rintel. 2020. Classification of Functional Attention in Video Meetings. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831>

- 3376546
- [16] Kate Lister. 2020. *Work-At-Home After Covid-19—Our Forecast*. Global Workplace Analytics. <https://globalworkplaceanalytics.com/work-at-home-after-covid-19-our-forecast>
 - [17] Logitech. 2020. *Logitech MeetUp and Rally ConferenceCams Enable Hybrid Meetings for Radisson Customers*. Logitech. <https://www.logitech.com/en-ch/video-collaboration/resources/case-study/radisson-hotel-group.html>
 - [18] Riëks op den Akker, Dennis Hofs, Hendri Hondorp, Harm op den Akker, Job Zwiers, and Anton Nijholt. 2009. Supporting engagement and floor control in hybrid meetings. In *Cross-Modal Analysis of Speech, Gestures, Gaze and Facial Expressions*. Springer, Berlin, Heidelberg, 276–290. https://doi.org/10.1007/978-3-642-03320-9_26
 - [19] Irwin Pollack and J. M. Pickett. 1957. Cocktail Party Effect. *The Journal of the Acoustical Society of America* 29, 11 (Nov. 1957), 1262–1262. <https://doi.org/10.1121/1.1919140>
 - [20] A. Raake and C. Schlegel. 2008. Auditory assessment of conversational speech quality of traditional and spatialized teleconferences. In *ITG Conference on Voice Communication [8. ITG-Fachtagung]*. VDE, Berlin, 1–4.
 - [21] Alexander Raake and Hagen Wierstorf. 2020. Binaural Evaluation of Sound Quality and Quality of Experience. In *The Technology of Binaural Understanding*, Jens Blauert and Jonas Braasch (Eds.). Springer International Publishing, Cham, 393–434. https://doi.org/10.1007/978-3-030-00386-9_14
 - [22] Banu Saatçi, Kaya Akyüz, Sean Rintel, and Clemens Nylandsted Klokmoose. 2020. (Re) Configuring Hybrid Meetings: Moving from User-Centered Design to Meeting-Centered Design. *Computer Supported Cooperative Work (CSCW)* 29, 6 (2020), 769–794. <https://doi.org/10.1007/s10606-020-09385-x>
 - [23] Banu Saatçi, Roman Rädle, Sean Rintel, Kenton O'Hara, and Clemens Nylandsted Klokmoose. 2019. Hybrid Meetings in the Modern Workplace: Stories of Success and Failure. In *International Conference on Collaboration and Technology*. Springer, Cham, 45–61. https://doi.org/10.1007/978-3-030-28011-6_4
 - [24] Ashutosh Saxena and Andrew Y Ng. 2009. Learning sound location from a single microphone. In *2009 IEEE International Conference on Robotics and Automation*. IEEE, Kobe, Japan, 1737–1742. <https://doi.org/10.1109/ROBOT.2009.5152861>
 - [25] Janto Skowronek, Falk Schiffner, and Alexander Raake. 2013. On the influence of involvement on the quality of multiparty conferencing. In *4th International Workshop on Perceptual Quality of Systems*. ISCA, Vienna, 25–30. <https://doi.org/10.21437/pqs.2013-25>
 - [26] Carole B Sox, Tena B Crews, and Sheryl F Kline. 2014. Virtual and hybrid meetings for generation X: using the Delphi method to determine best practices, opportunities, and barriers. In *Journal of Convention & Event Tourism*, Vol. 15:2. Routledge, UK, 150–169. <https://doi.org/10.1080/15470148.2014.896231>
 - [27] Anthony Tang, Michel Pahud, Kori Inkpen, Hrvoje Benko, John C. Tang, and Bill Buxton. 2010. Three's Company: Understanding Communication Channels in Three-Way Distributed Collaboration. In *Proceedings of the 2010 ACM Conference on Computer Supported Cooperative Work (Savannah, Georgia, USA) (CSCW '10)*. Association for Computing Machinery, New York, NY, USA, 271–280. <https://doi.org/10.1145/1718918.1718969>
 - [28] J. Valin. 2018. A hybrid DSP/deep learning approach to real-time full-band speech enhancement. In *2018 IEEE 20th international workshop on multimedia signal processing (MMSP)*. IEEE, Vancouver, BC, 1–5. <https://doi.org/10.1109/MMSP.2018.8547084>
 - [29] Bin Xu, Jason Ellis, and Thomas Erickson. 2017. Attention from Afar: Simulating the Gazes of Remote Participants in Hybrid Meetings. In *Proceedings of the 2017 Conference on Designing Interactive Systems (Edinburgh, United Kingdom) (DIS '17)*. Association for Computing Machinery, New York, NY, USA, 101–113. <https://doi.org/10.1145/3064663.3064720>
 - [30] Y. Xu, J. Du, L. Dai, and C. Lee. 2015. A Regression Approach to Speech Enhancement Based on Deep Neural Networks. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 23, 1 (2015), 7–19. <https://doi.org/10.1109/TASLP.2014.2364452>
 - [31] Nicole Yankelovich, Jonathan Kaplan, Joe Provino, Mike Wessler, and Joan Morris DiMicco. 2006. Improving Audio Conferencing: Are Two Ears Better than One?. In *Proceedings of the 2006 20th Anniversary Conference on Computer Supported Cooperative Work (Banff, Alberta, Canada) (CSCW '06)*. Association for Computing Machinery, New York, NY, USA, 333–342. <https://doi.org/10.1145/1180875.1180926>
 - [32] Nicole Yankelovich, William Walker, Patricia Roberts, Mike Wessler, Jonathan Kaplan, and Joe Provino. 2004. Meeting Central: Making Distributed Meetings More Effective. In *Proceedings of the 2004 ACM Conference on Computer Supported Cooperative Work (Chicago, Illinois, USA) (CSCW '04)*. Association for Computing Machinery, New York, NY, USA, 419–428. <https://doi.org/10.1145/1031607.1031678>